

Mandatory assignment for Term Structure Models

1 Foundational models

Consider a probability measure $(\Omega, \mathcal{F}, \mathbb{Q})$ with filtration $\{\mathcal{F}_t\}_{0 \leq t \leq T^*}$. We will consider the two processes X_t and Y_t described by the SDEs

$$\begin{aligned} \frac{dX_t}{X_t} &= \sigma_X(t) dW_t^{\mathbb{Q}}, & X_0 &= x, \\ dY_t &= \sigma_Y(t) dW_t^{\mathbb{Q}}, & Y_0 &= x, \end{aligned}$$

for deterministic and strictly positive functions $\sigma_X(\cdot)$ and $\sigma_Y(\cdot)$ (both in $\mathcal{L}^2$), and where $x > 0$.

1. Derive the solutions to X_t and Y_t . What is the distribution of X_t and Y_t for $t > 0$? Is X_t a martingale? Is Y_t ?
2. Now assume that the volatility functions are constant, such that $\sigma_X = 0.2$ and $\sigma_Y = 4$ and $x = 20$. Plot the density of X_t and Y_t for $t = 5$.
3. For general (so no longer constant!) volatility functions again, derive closed-form solutions to the expectations below:

$$\mathbb{E}^{\mathbb{Q}} \left\{ (X_T - K)^+ \right\} \quad \text{and} \quad \mathbb{E}^{\mathbb{Q}} \left\{ (Y_T - K)^+ \right\}.$$

4. Fully ignore the drift and focus only on the diffusion. Based on the qualitative differences between the processes X_t and Y_t , discuss how appropriate X_t and Y_t are as models for:
 1. The short rate r_t .
 2. A zero-coupon bond $P(t, T)$, for a fixed $T > t$.
 3. A Libor rate $L(t, T, T + \Delta)$.
 4. A Libor spread $L(t, T, T + \Delta_1) - L(t, T, T + \Delta_2)$, for $\Delta_2 > \Delta_1 > 0$.
 5. A stock price.

2 A stochastic volatility term structure model

Consider a probability measure $(\Omega, \mathcal{F}, \mathbb{Q})$ with filtration $\{\mathcal{F}_t\}_{0 \leq t \leq T^*}$. We will assume that the dynamics of the short rate $r(t)$ under the risk-neutral measure $\mathbb{Q}$ is given by the stochastic differential equation

$$dr(t) = \kappa_r (\theta_r - r(t)) dt + \sigma_r \sqrt{\alpha + \beta v(t)} dW_1^{\mathbb{Q}}(t) \quad (1)$$

$$dv(t) = \kappa_v (\theta_v - v(t)) dt + \sigma_v \sqrt{\alpha + \beta v(t)} dW_2^{\mathbb{Q}}(t), \quad (2)$$

with $d\langle W_1^{\mathbb{Q}}, W_2^{\mathbb{Q}} \rangle_t = \rho dt$. We will denote the combined system by $\mathbf{X}_t = (r(t), v(t))^{\top}$. Unless explicitly stated so, you can assume that the parameters of the model are given as reported in Table 1. The same goes for the initial state of the system $\mathbf{X}_t$ (or sometimes $\mathbf{X}_0$), which will be equal to the long-term mean θ unless explicitly stated otherwise.

1. Rewrite the model into its affine characteristics (K_0, K_1, H_0, H_1) such that the drift and instantaneous variance is given by

$$\mu(x) = K_0 + K_1 x \quad (\sigma(x)\sigma(x)^{\top})_{ij} = H_{0ij} + H_{1ij} \cdot x.$$

Table 1: Model parameters

κ_r	κ_v	θ_r	θ_v	σ_r	σ_v	α	β	ρ
0.5	1.2	0.05	1	0.08	1.0	0.5	0.5	0.25

2. Derive the Ricatti equations for a zero-coupon bond $P(t, T)$ in the model above, such that

$$P(t, T) = e^{\alpha_z(T-t) + \beta_z(T-t) \cdot \mathbf{X}_t}.$$

Solve the ODEs (numerically), and plot the values of $\alpha_z(\cdot)$ and $\beta_z(\cdot)$, the discount curve $\{P(t, t + \Delta)\}_{0 \leq \Delta \leq 30}$ and the yield curve $\{y(t, t + \Delta)\}_{0 \leq \Delta \leq 30}$ for the model.

3. We will now verify that the ZCB formula derived above is correct using Monte Carlo. Simulate the model using e.g., the Euler-Maruyama scheme, from time $t = 0$ to $T = 30$ and compute the value of $P(0, T)$ for $0 \leq T \leq 30$. Use at least 10 000 sample paths. Plot both the analytical solutions and the Monte Carlo approximations in the same plot. Describe your simulation scheme.
4. Use your simulation scheme developed above to simulate samples of $r_T | \mathbf{X}_0$ with $T = 5$. Plot the (sample/empirical) CDF of $r_T | \mathbf{X}_0$. Are negative interest rates possible/probable in this model? Furthermore, plot the sample CDF of $P(T, T + \Delta)$ for Δ equal to one and ten years in the same manner. Comment on the difference between these two CDF's.
5. Consider the transform

$$\psi(y; \mathbf{X}_t, t, \Delta) \triangleq \mathbb{E}_t^{\mathbb{Q}} \left\{ \mathbf{1}_{\beta_z(\Delta) \cdot \mathbf{X}_T \leq \log(y) - \alpha_z(\Delta)} \right\}, \quad (3)$$

where $\alpha_z(\Delta)$ and $\beta_z(\Delta)$ are the solutions to the ZCB formula above. Describe how to compute this function using transform methods. Compute and plot $\{\psi(y; \mathbf{X}_t, t, \Delta)\}$ for $y \in [0.5, 1.5]$ for both $\Delta = 1$ and $\Delta = 10$. What is ψ ?

Hint: This is a specific form of the more general G -transform in (4).

6. Now consider the full G -transform given by

$$G_{a,b}(y) = G_{a,b}(y; \mathbf{X}_t, t, T, \Delta) \triangleq \mathbb{E}_t^{\mathbb{Q}} \left\{ e^{-\int_t^T (\rho_0 + \rho_1 \cdot \mathbf{X}_u) du} e^{a \cdot \mathbf{X}_T} \mathbf{1}_{b \cdot \mathbf{X}_T \leq y} \right\}, \quad (4)$$

where we fix $\rho_0 = 0$ and $\rho_1 = (1, 0)^\top$. Explain what the terms

$$G_1(x) = G_{0, \beta_z(\Delta)} [\log(x) - \alpha_z(\Delta)] \quad \text{and} \quad G_2(x) = e^{\alpha_z(\Delta)} G_{\beta_z(\Delta), \beta_z(\Delta)} [\log(x) - \alpha_z(\Delta)]$$

represent. Compute the values of $G_1(x)$ and $G_2(x)$ for $T = 5$ and $\Delta = 1$ and where $x = \frac{P(t, T + \Delta)}{P(t, T)}$. Describe how to evaluate G_1 and G_2 using Monte Carlo methods and verify your solutions using Monte Carlo.

7. Zero-coupon bond call (ZCC) and put (ZCP) options are given by

$$\begin{aligned} \text{ZCC}(K, t, T, \Delta) &= \mathbb{E}_t^{\mathbb{Q}} \left\{ e^{-\int_t^T r_u du} [P(T, T + \Delta) - K]^+ \right\} \\ \text{ZCP}(K, t, T, \Delta) &= \mathbb{E}_t^{\mathbb{Q}} \left\{ e^{-\int_t^T r_u du} [K - P(T, T + \Delta)]^+ \right\} \end{aligned}$$

Describe how to price these using transform methods.

Now assume that $t = 0$, $T = 5$ and $\Delta = 1$. For $K = 0.97$ compute and report the value of a call and a put option, and (of course) verify your results using Monte Carlo. Compute the difference

$$ZCC - ZCP,$$

and compare to the value $P(t, T + \Delta) - P(t, T)K$. What is going on here?

8. Consider the two series

$$k_i^p = K^* - d \left(1 - \frac{i}{n}\right) \quad k_i^c = K^* + d \frac{i}{n}$$

for $i = 0, \dots, n$, and with $K^* = \frac{P(t, T + \Delta)}{P(t, T)}$ and $d = 0.2$. Using $n = 10$, compute the value of puts with strikes k_i^p and calls with strikes k_i^c . Compute the implied volatility of these options and produce a plot with strikes on the x -axis and implied volatilities on the y -axis. Remember discounting.

Do the same as above, except now for the two models given by:

1. everything is the same as in Table 1, except $\sigma_v = 2$.
2. everything is the same as in Table 1, except $\rho = -0.25$.

Describe how the volatility smile changes depending on the parameters.

3 A multifactor affine model

Consider the two-dimensional affine model with dynamics under the risk-neutral measure $\mathbb{Q}$ given by

$$\begin{aligned} dr(t) &= \kappa_r (\theta - r(t)) dt + \sigma_r \sqrt{v(t)} dW_1^{\mathbb{Q}}(t), \\ dv(t) &= \kappa_v (1 - v(t)) dt + \sigma_v \sqrt{v(t)} dW_2^{\mathbb{Q}}(t), \end{aligned}$$

where $W_1^{\mathbb{Q}}(t)$ and $W_2^{\mathbb{Q}}(t)$ are standard (uncorrelated) Brownian motions, $\kappa_r, \kappa_v, \sigma_r$ and σ_v are strictly positive parameters, θ is an unbounded parameter. Notice that this model is a specific instance of the previous model.

Table 2: Model parameters

κ_r	κ_v	θ_r	σ_r	σ_v	γ_1	γ_2
0.5	1.2	0.05	1	0.08	0.1	0.2

1. Rewrite the process into the form

$$d\mathbf{X}(t) = K(\mu - \mathbf{X}(t))dt + \Sigma S(\mathbf{X}(t))dW^{\mathbb{Q}}(t), \quad (5)$$

where $\mathbf{X}(t) = (r(t), v(t))^{\top}$.

2. Derive the dynamics of the model under the physical measure for both the completely and essentially affine market price of risk specification. How do they differ?

- Using the completely affine market price of risk specification, compute the matrix $\mathbf{A}$ and vector $\mathbf{b}$ such that

$$\mathbb{E}^{\mathbb{P}} \{ \mathbf{X}(t + \Delta) | \mathcal{F}_t \} = \mathbf{A} \mathbf{X}(t) + \mathbf{b},$$

for a given timestep $\Delta > 0$. What is the unconditional mean of $\mathbf{X}(t)$?

- Compute (and report) the conditional covariance matrix of $\mathbf{X}(t + \Delta) | \mathbf{X}(t)$ using the formula from Christensen et al. (2015) and with parameters as in Table 2 for $\Delta = 1/52$ and with $\mathbf{X}(t)$ equal to its long-term mean. Also compute and report the unconditional covariance matrix $\mathbb{V}^{\mathbb{P}} \{ \mathbf{X}(t) \}$.
- Describe the issue with this model with respect to the Kalman filter, and how one could adapt the method.

Hint: If we have a variable X with finite mean and variance, then we can approximate it as a Gaussian r.v. Y by moment matching.

- We now wish to estimate the dual-measure model with $\mathbb{Q}$ -dynamics (5) and $\mathbb{P}$ -dynamics from the complete affine market price of risk, such that the market price of risk is fully determined by a 2-dimensional vector γ . We do so using Kalman filtering in conjunction with the prediction error decomposition. The dataset `data.csv` on Absalon contains a time series of weekly ($\Delta = 1/52$) measurements of yields with times to maturity

1, 2, 3, 4, 5, 10 and 20 years.

All measurements have been corrupted with zero-mean uncorrelated Gaussian measurement noise with covariance s_r^2 . The combined parameters are thus given by

$$\Theta = (\kappa_r, \kappa_\sigma, \theta, \sigma_r, \sigma_v, \gamma_1, \gamma_2, s_r)$$

Derive the measurement and transition equations for the model, and implement the Kalman filter. Estimate the parameter using the prediction error decomposition and report your maximum likelihood estimate of Θ and the corresponding loglikelihood at optimum. Plot the time series of updated (*a posteriori*) states.

- You can use e.g., `optim` in R or Scipy's `optimize.minimize` for finding the maximum likelihood estimate. Common optimization algorithms for solving these kinds of problems include Nelder-Mead or L-BFGS-B, but you're free to use any method you like.
- You should initialize the algorithm at the unconditional mean and covariance of $\mathbf{X}(t)$.

References

Christensen, J. H., Lopez, J. A., and Rudebusch, G. D. (2015). Analytical formulas for the second moment in affine models with stochastic volatility. Technical report.